Electrostatics

Question 1

Two identical metal spheres are charged, one positively and one negatively, and then brought into contact and separated. What happens to their charges? Explain.

Question 2

Two point charges q1 = +3.0 μC and q2 = -2.0 μC are separated by 0.50 m. Find the magnitude and direction of the force.

Capacitance and DC Circuits

Question 1

In a series circuit, what happens to the current and voltage across each resistor if one resistor is increased? Explain.

Question 2

A 12 V battery is connected to 2Ω, 4Ω, and 6Ω resistors in series. Find total resistance, current, and voltage across the 4Ω resistor.

Magnetism

Question 1

A charged particle moves parallel to a uniform magnetic field. What magnetic force does it experience? Explain.

Question 2

A proton moves at 2.0×10^6 m/s perpendicular to a 0.30 T magnetic field. Find the magnetic force.

Electrical Induction and AC Circuits

Question 1

According to Lenz’s Law, why does an induced current oppose the change that produced it?

Question 2

A coil with 200 turns experiences a change in magnetic flux of 0.015 Wb in 0.20 s. What is the induced emf?

Electromagnetic Waves and Wave Optics

Question 1

Why can electromagnetic waves travel through a vacuum while mechanical waves cannot?

Question 2

Light of wavelength 600 nm passes through slits separated by 0.20 mm. Screen is 2.0 m away. Find fringe spacing.

Geometric Optics

Question 1

Why does a converging lens form a real image for distant objects but a virtual image when inside the focal length?

Question 2

An object is 30 cm from a lens with focal length 10 cm. Find image distance and magnification.

Quantum and Atomic Physics

Question 1

Why does increasing light intensity not increase kinetic energy in the photoelectric effect?

Question 2

Light frequency 1.2×10^15 Hz hits a metal with work function 4.0 eV. Find maximum kinetic energy.

Nuclear Physics

Question 1

Why are some nuclei unstable and undergo radioactive decay?

Question 2

A sample with half-life 10 days starts at 80 g. How much remains after 30 days?